

AYUSH KUMAR BHADANI

+91-9122883316 ✉ ayushbhadani0915@gmail.com

🌐 [LinkedIn Profile](#) 🐙 [GitHub](#) 🏆 [LeetCode](#)

Education

Bangalore Institute of Technology

2023-2027

Bachelor of Technology in Computer Science and Engineering (Data Science) | **CGPA: 9.45**

Bangalore, Karnataka

Delhi Public School

2022

Senior Secondary Education in Science (PCM) | **Percent: 87.2**

Ranchi, Jharkhand

St. Michael's School

2020

Secondary Education | **Percent: 95.6**

Ranchi, Jharkhand

Relevant Coursework

- Data Structures and Algorithms
- Object-Oriented Programming
- Computer Networks
- Database Management Systems
- Operating Systems
- Machine Learning

Projects

CodeVerdict | Python, FastAPI, Groq/Llama 3, FAISS, RAG, Agentic AI

Live App

Designed, built, and orchestrated a **multi-agent AI workflow** with **4 specialized async agents** (Security, Quality, Performance, Test Coverage) using Python **asyncio** as the **agentic AI framework**, enabling concurrent execution with isolated error handling per agent.

Developed a **Retrieval-Augmented Generation (RAG)** system using **Sentence Transformer** embeddings, **FAISS** vector search, and **prompt engineering** to ground LLM agent outputs in secure-coding reference documentation, processing **unstructured text** (code files, PRs, documentation).

Integrated **Bandit static analysis (SAST)** into the Security Agent's **LLM** pipeline with a deterministic merge-verification step, ensuring static analysis findings are never omitted from the final synthesized output.

Designed and ran a **precision/recall evaluation suite** against a labeled PR benchmark dataset, achieving **87.5% precision** and **100% recall**, with documented failure-mode analysis and a tracked mitigation plan.

AskSQL | Python, FastAPI, DuckDB, Groq/Llama 3, AST Validation

Live App

Built **AskSQL**, a natural-language-to-SQL engine that lets users upload any dataset and query it in plain English, using a custom **zero-dependency token-matching retrieval algorithm** in place of vector embeddings, removing the need for ChromaDB or Sentence Transformers.

Re-architected the backend around **DuckDB's native streaming CSV parser** in place of Pandas, cutting baseline runtime memory to **~60MB** and enabling stable deployment on free-tier infrastructure.

Designed a **multi-layer SQL safety system** combining **AST-based query validation** with per-session sandbox isolation, blocking destructive operations and cross-session data leakage on untrusted, user-uploaded datasets.

Evaluated the pipeline on a **20-question benchmark** across simple, medium, and complex query tiers, achieving **60% execution accuracy** with an automated retry-and-revalidation loop, at an average end-to-end latency of **~1.27s**.

Technical Skills

Languages: Python, C++, SQL

AI/ML & NLP: scikit-learn, Sentence Transformers, K-Means, RAG, Prompt Engineering

Backend & APIs: FastAPI, asyncio, LLM APIs (Groq, Llama 3), REST APIs

Databases & Tools: MongoDB, MySQL, DuckDB, FAISS, Git, GitHub

Data & Visualization: Pandas, NumPy, Plotly, Power BI

Methodologies: Agile/Scrum, EDA

Soft Skills: Analytical Thinking, Problem Solving, Communication, Leadership

Achievements

VTU Branch Topper - 3rd & 6th Semester | CGPA: 9.86, 10.0

Secured the highest academic performance in the **CSE (Data Science)** branch at Bangalore Institute of Technology, achieving a perfect **10.0 CGPA** in the 6th semester.

Core Technical Member | Google Developer Group (GDG)

Collaborated with a **5-member team** to deliver the official GDG website across sprints; conducted **3+ workshops** and **2+ regional events** reaching **1000+ participants**.

Leadership & Activities

Organized large-scale technical events including **ambITION 2.0** and **AI Study Jams** for **200+ participants**, handling planning, coordination, and execution.

Represented college at **inter-college hackathons** and mentored **10+ junior students** on **DSA** and Python fundamentals, fostering peer learning and collaborative problem-solving.